Introduction to T-SQL Capstone Project

Learning Outcomes

By completing this capstone project, students will demonstrate their ability to:

- Create and modify database tables using DDL statements.
- Insert, update and delete records using DML statements.
- Retrieve data using the SELECT statement.
- Apply column aliases, concatenation and COALESCE().
- Filter records using WHERE, DISTINCT, LIKE, IN, BETWEEN, IS NULL, AND, OR, and <>.
- Retrieve related data using SQL joins.
- Sort, group and aggregate data.
- Use SQL date and string functions.
- Create and use subqueries.
- Create a view.
- Create a Common Table Expression (CTE).
- Create and execute a stored procedure.
- Use IF...ELSE logic.
- Implement transactions.
- Handle errors using TRY...CATCH.

Business Scenario

SuperMart is opening a new branch in Johannesburg.

The company has hired you as a Junior Database Developer for their Data Cluster.

Before the system goes live, management wants you to build a small database and generate reports that will assist the sales department.

You are required to complete the following tasks.

Activity 1 – Create the Database

Create a database called SuperMart_Db

Create the following tables:

Customers table with the following columns:

- CustomerId
- FirstName
- LastName
- City
- Phone (n.b. this column is the only one that can have nulls)
- Email

Orders table with the following columns:

- OrderId
- CustomerId
- OrderDate
- StatusCode i.e P – Pending, D – Delivered, C – Canceled
- TotalAmount

Task Deliverables:

- Columns should not allow nulls unless stated so.
- Add Primary Keys in both the Customers and Orders table
- Create the foreign key between Customers and Orders
- Use of datatypes at your own discretion.

Activity 2 – Populate the Database

Insert

- 7 Customers
- 10 Orders

Requirements

- At least 3 countries, i.e Pretoria, Cape Town , Joburg
- Some NULL phone numbers
- Different order dates from 1 Jan 2026 to 31 Dec 2026
- Different order amounts
- Different order statuses
- At least two customers with no orders

Activity 3 – Basic Data Retrieval

The Sales Manager requires a customer contact report.

Write a **single T-SQL query** that displays the following information for every customer:

- Customer ID
- Customer's full name by concatenating **FirstName** and **LastName** into a column named **Customer Name**
- Country
- City
- Phone number, replacing any NULL values with **'No Phone Number'** using the COALESCE() function

Activity 4 – Filtering Data

A) *The Marketing Manager wants to identify customers from Southern Africa*. Write a single T-SQL query that displays all customers who are from **Gauteng**(i.e. Joburg and Pretoria) using the IN operator. Display only the customer's full name, email, and city.

B)*The Sales Manager needs a report of orders placed during the first quarter of 2026*.

Write a T-SQL query that displays all orders placed between **1 January 2026** and **31 March 2026** using the BETWEEN operator. Display the Order ID, Customer ID, Order Date, Status, and Total Amount.

Activity 5 – SQL Joins

Inner Join

The Sales Manager wants a report of customers who have placed orders.

Write a T-SQL query that displays the following information using an **INNER JOIN**:

- Customer Name
- Order ID
- Order Date
- Total Amount

Left Join

The Customer Relations department wants to identify customers who have not yet placed an order.

Write a T-SQL query that displays **all customers**, including those who have not placed any orders, using a **LEFT JOIN**. Display:

- Customer Name
- Order ID
- Order Date
- Total Amount

Right Join

The Operations Manager wants to review every order in the system, including orders that may not have matching customer records.

Write a T-SQL query using a **RIGHT JOIN** that displays:

- Customer Name
- Order ID
- Order Date
- Total Amount

Full Outer Join

The Database Administrator is performing a data integrity audit.

Write a T-SQL query using a **FULL JOIN** that displays:

- Customer Name
- Order ID
- Order Date
- Total Amount

Activity 6 – Sorting, Aggregation, Date and String Functions

Task 1

The Customer Relations Manager requires a customer directory.

Write a T-SQL query that displays the following information for all customers:

- Customer Name in **UPPERCASE**
- Country
- The length of the customer's first name
- Sort the results alphabetically by the customer's first name in ascending order.

Task 2

The Sales Manager wants to analyse customer distribution across different countries.

Write a T-SQL query that displays:

- Country
- Total number of customers in each country

Sort the results by the number of customers in descending order.

Task 3

The Finance department requires an order summary report.

Write a single T-SQL query that displays:

- The total number of orders
- The average order amount
- The highest order amount
- The lowest order amount

Task 4

The Operations Manager wants to review order activity.

Write a T-SQL query that displays:

- Order ID
- Order Date
- Order Year
- Order Month
- Number of days since the order was placed

Also, sort the results from the **highest order amount to the lowest**.

Activity 7 – Advance Queries and Stored Procedures

Section A

The Sales Manager wants to analyse customer purchasing activity.

Write two T-SQL queries that display the names of customers who **have placed at least one order** using a **subquery**.

Display the following columns:

- Customer ID
- Customer Name
- Country

Section B

Management requires a reusable report showing customer order activity.

Complete the following tasks:

1. Create a **View** named **CustomerOrders** that displays:
    - Customer Name
    - Order Date

- Total Amount

2. Create a **Common Table Expression (CTE)** that calculates the total number of orders placed by each customer and displays:
    - Customer Name
    - Number of Orders

Section C

The Customer Service department frequently needs to retrieve all orders placed by a specific customer without repeatedly writing the same query.

Create a **stored procedure** named **GetCustomerOrders** that accepts a **CustomerID** as an input parameter and returns the following information for that customer:

- Order ID
- Order Date
- Order Status
- Total Amount

After creating the stored procedure, execute it for **CustomerID = 1** to demonstrate that it works correctly.

Project Submission

Students must submit the following:

1. **GitHub Repository**
    - Create a public GitHub repository named **SuperMart-TSQL-Capstone**.
    - Push a single, well-structured .sql script containing:
        - Database creation
        - Table creation

    - Sample data insertion
    - All SQL queries
    - Views
    - Common Table Expressions (CTEs)
    - Stored procedures
    - Transactions
    - Error handling
- Ensure the repository includes meaningful commit messages that reflect your development progress.

2. **Execution Report**
    - Submit a Microsoft Word document containing screenshots demonstrating the successful execution of each section of the project in SQL Server Management Studio (SSMS).
    - Each screenshot must be clearly labelled with the corresponding section (e.g., *Part 1 – Database Creation*, *Part 4 – Filtering Data*).
3. **Code Documentation**
    - Ensure your SQL script is well-documented by including clear comments that explain the purpose of each section and major SQL statement.